Cognitive Diagnosis-Based Personalized Exercise Group Assembly via a Multi-Objective Evolutionary Algorithm

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***Abstract*—Exercise group recommendation plays an important role in many intelligent education tasks. However, existing approaches make recommendations based on the intrinsic features of exercises without considering students' learning abilities, or make selections from several pre-built exercise groups at the expense of flexibility. Furthermore, although many cognitive diagnosis approaches have successfully revealed students' abilities, how to leverage the diagnosis results for exercise group recommendation is hardly explored. To flexibly recommend suitable exercise groups to students, this paper proposes to assemble personalized exercises as a group based on students' abilities, called Personalized Exercise Group Assembly (PEGA). To solve the PEGA task, we first formulate it as a constrained multi-objective problem (CMOP), where three objectives are designed for enabling the assembled exercises to enhance students' abilities on both weak and new knowledge concepts. Then, we devise an extended neural cognitive diagnosis model to learn student's ability/proficiency on all knowledge concepts to compute the weakness consolidation objective. Besides, we propose a dual-encoding and dual-population based co-evolutionary algorithm to tackle the CMOP, where the main population with binary encoding is used to search which exercises are selected, and the auxiliary population with integer encoding is responsible for accelerating the convergence of the main population via guiding offspring generation of the main population. Experiments on three popular datasets demonstrate the effectiveness of exercises assembled by the proposed algorithm compared to state-of-the-art exercise group recommendation approaches, where our assembled exercises can enhance students' proficiency on both poorly mastered and new knowledge concepts.**

***Index Terms*—Exercise group recommendation, assemble exercises, personalized exercise group assembly, multi-objective problem, co-evolutionary algorithm.**

I. Introduction

EXERCISE group recommendation usually selects m exercises from a large exercise pool as a group for each student to improve the student's ability/proficiency on the studied knowledge concepts and new knowledge concepts as much as possible, which can make many education tasks (e.g., homework assignment [1] and test paper generation [2]) more intelligent. Compared to traditional unified exercising mode that assigns a unified exercise group to all students in the same class, the exercise group recommendation can not only improve students' learning efficiencies and decrease their learning burdens but also enable each student to learn and practice on their own pace, which can simulate students' interests in learning [3] to some extent.

Up to now, there have been some exercise group recommendation approaches, which can be mainly classified into two categories. The first type of approaches [1], [2] commonly recommends a series of exercises as a group to a student mainly based on the intrinsic characteristics of exercises (e.g, exercise difficulty and distinction) without considering students' personalized learning abilities, which leads to the fact that the recommended exercise groups are usually of little avail for the students with strong learning abilities but a bit difficult for the students with weak knowledge proficiency. The second type of approaches commonly provides multiple pre-built exercise groups with different difficulty levels (e.g., easy, medium, difficult), which are constructed by education experts. Then, a simple rule is utilized to adaptively select a suitable exercise group from pre-built exercise groups for each student according to the student's integrated ability level (e.g., fine, medium, or to be improving). The second type has been adopted by some famous adaptive learning systems (e.g., IFLYTEK AI Learning Machine

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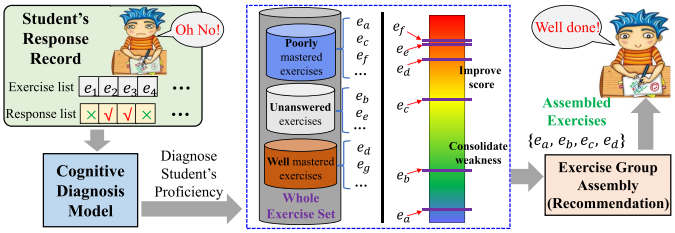


Fig. 1. The illustration of personalized exercise group assembly. Based on the student's proficiency obtained by a cognitive diagnosis model, the PEGA will employ an exercise group assembly strategy to recommend a group of personalized exercises to the student for improving the student's ability.

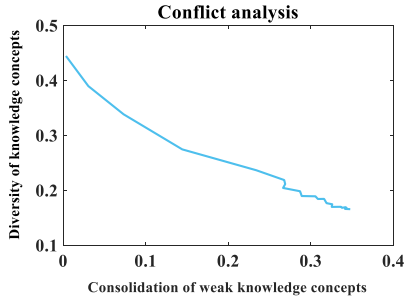


Fig. 2. Illustration of conflict analysis for the first and third objectives. It is intuitive that when most exercises in an assembled exercise group are used for consolidating student's weak concepts, there will be fewer exercises containing new concepts, thus the diversity of concepts naturally will be poor.

and Tencent Education Application Platform[1]). However, due to the poor flexibility of the pre-built exercise groups, the recommended exercise group selected from pre-built exercise groups may not be the most suitable for each student.

To achieve the flexible exercise group recommendation suitable to each student, this paper proposes the Personalized Exercise Group Assembly (PEGA) task that adaptively assembles personalized exercises as a group for each student concerning the student's learning ability. Note that cognitive diagnosis (CD) models [4], [5], [6], [7] are employed here to accurately diagnose students' learning abilities based on their response records, where their diagnosed results have been less explored in the previous exercise group recommendation. For better understanding the PEGA task, Fig. 1 summarizes its main procedure. Firstly, a CD model is used to diagnose the student's ability/proficiency on all knowledge concepts based on the student's response records. According to the diagnosed results, then an exercise group assembly (recommendation) strategy is used to assemble a group of personalized exercises for the student. Although the proposed PEGA task is a special case of exercise group recommendation, there exist two important characteristics for the PEGA when assembling a group of exercises, that is, not only each student's personalized proficiency is considered but also the recommended exercise group is built by flexibly assembling different exercises instead of selecting a candidate from pre-built exercise groups.

To tackle the PEGA task well, we formulate it as a constrained multi-objective optimization problem (CMOP) with three objectives: the weakness consolidation objective, the new knowledge exploration objective, and the knowledge diversity objective. However, as shown in Fig. 2, there exists a conflict between the first objective and the third objective, where fewer new concepts will be included and the diversity of concepts will be poor when more recommended exercises focus on consolidating the student's weak concepts. Moreover, there are other two challenges to be overcome when solving the CMOP. First, it is challenging to obtain accurate students' proficiency on all concepts for computing the first objective since it is difficult for existing CD models to learn good diagnosis results on the concepts that students have not done the associated exercises. The second challenge is the fact that the formulated CMOP is a constrained sparse large-scale MOP. As demonstrated in [8], there is no multi-objective evolutionary algorithm (MOEA) devised for constrained sparse large-scale MOPs and existing MOEAs cannot solve them well.

To this end, we suggest an effective CD model for obtaining accurate student's proficiency, and propose a co-evolutionary algorithm to tackle the CMOP, where a novel genetic operator and the cooperation of two populations are to settle the difficulties caused by the sparsity and constraints of the CMOP, respectively. Specifically, our main contributions are as follows:

1) We are the first to propose the concept of the PEGA and devise three objectives to ensure the good quality of the assembled exercise group, including weakness consolidation, new knowledge exploration, and knowledge diversity. The first objective is to make students' weak knowledge concepts benefit from the assembled exercise group as much as possible, the last two objectives are responsible for maintaining more new knowledge concepts and ensuring the diversity of knowledge concepts, respectively, in the assembled exercise group. By doing so, we formally formulate the PEGA task as a CMOP containing four constraints, where the first two constraints are caused by the intrinsic nature of the problem and the other two constraints are used to make the student maintain high learning enthusiasm.
2) To overcome two challenges in the CMOP, we first devise an extended neural CD model (termed ENCD) based on NeuralCD [6] to obtain the student's accurate proficiency on all knowledge concepts for computing the weakness consolidation objective. Then, a **co**-**e**volutionary algorithm based on **d**ual-en**c**oding and **d**ual-**p**opulation (CoEDCDP) is proposed to solve the CMOP. Specifically, a main population with binary encoding is used to search which exercises are selected, and an auxiliary population with integer encoding is responsible for accelerating the convergence of the main population by a novel genetic operator guiding the offspring generation of the main population. By doing so, the novel genetic operator and the cooperation of two populations can overcome the difficulties caused by the sparsity and constraints of the CMOP.

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3) Experimental results on three widely used datasets, namely ASSISTments, JunYi and SLP, demonstrate the effectiveness of exercises assembled by the proposed CoE-DCDP compared with state-of-the-art exercise group recommendation approaches. Besides, it is verified that our assembled exercise groups can enhance students' proficiency on not only poorly mastered knowledge concepts but also new never saw knowledge concepts.

The rest of the paper is organized as follows. Section II will present the background and the related work, and Section III gives the mathematical formulation of the PEGA task and proposes the extended neural CD model. The proposed approach is elaborated in Section IV, followed by the empirical evaluations presented in Section V. Finally, conclusions are drawn in Section VI.

II. Background and Related Work

A. The PEGA Task

The more relevant tasks to our work are homework assignment [1] and test paper generation [2]. The traditional approaches mainly focus on the exercise features, e.g., difficulty and distinction, and do not consider customized homework or test with respect to student's abilities. In recent several adaptive learning systems (e.g., IFLYTEK AI Learning Machine), the homework or test papers with different difficulty levels are constructed by education experts. A simple rule is made to match the students with the homework/test paper according to their integrated ability levels. Different from these existing work, we attempt to auto-generate a homework or test paper for each student according to their detailed knowledge proficiency.

Personalized exercise recommendation with respect to the student's learning ability is also related to our work. The main difference among them is that the former just considers the benefit of single exercise while the latter considers the portfolio benefits of m exercises. In addition, the traditional exercise recommendation approaches mainly focus on making decisions following one goal. For example, some existing studies [3], [9], [10], [11] try to discover the weakness of students and then make decisions of recommending their non-mastered exercises (i.e., the goal is to address the weakness of students); Some others [12], [13], [14], [15] attempt to obtain students' learning abilities and the difficulty of exercises or knowledge concepts and then make decisions of recommending the exercises whose difficulty is suitable for the students (i.e., the goal is to maintain the learning enthusiasm of students).

In recent years, some researchers have pointed out that making recommending decisions of simply considering one goal is reasonable but ignores the long-term learning needs of students in practice [14], [16]. For example, it is not optimal that the exercise recommender system would tend to keep pushing the student to practice non-mastered exercises including even the ones that the student cannot deal with. Almost these studies firstly design the task-oriented optimization objectives and then linearly merge these objectives with manually-designed weight as a single objective optimization problem. For example, Huang et al. [14] proposed a novel deep reinforcement learning framework for recommending exercise learning path, which linearly merges three objectives (i.e., Review & Explore, Smoothness of difficulty level and Engagement) as the reward function. Li et al. [12] defined five objectives (e.g., the difference of the student's learning ability and difficulty of learning materials) for extracurricular learning materials recommendation and linearly merged these as an optimization objective, and then exploited an evolutionary algorithm named FBPSOS to search the suitable learning materials for each student. However, how to make strategies and design optimization objectives for adaptively recommending m exercises from a larger exercise bank is less explored. Moreover, these single-objective optimization methods are not suitable for such complex combinatorial optimization problems.

B. Cognitive Diagnosis

In intelligent education systems, the purpose of CD is to diagnose the students' proficiency on specific knowledge concepts according to the students' response records. In recent years, there have been a lot of CD approaches proposed, such as the representatives of traditional CD approaches DINA [4], and MIRT [5], and the representatives of deep learning-based CD approaches NeuralCDM [6] and DIRT [7].

However, there exist challenges for these CD approaches to discover the student's learning ability over knowledge concepts that are not related to his/her response records. Although there have been RCD [17] and CDGK [18] attempting to address this issue by incorporating the prior prerequisite relations between knowledge concepts, the annotation of the relations is labor-intensive and costly, which is difficult for most online learning systems to obtain this information. Therefore, it is necessary to design an effective and efficient CD model, which can diagnose the student's proficiency on all knowledge concepts merely based on the student's response logs. By doing so, the diagnosis results can be used to compute the first proposed objective for the PEGA task.

C. Large-Scale Evolutionary Multi-Objective Optimization

As summarized in [8], existing MOEAs solve large-scale multi-objective optimization problems (LSMOPs) mainly from three perspectives.

The first type of MOEAs is based on decision variable grouping, which divides the decision variables into several groups, and then optimizes each group of variables alternately. The pioneer work CCGDE [19] proposed by Antonio and Coello leverages a random group technique and a co-evolutionary framework to solve LSMOPs. Similar random grouping-based MOEAs include OD-NSGA [20] and MOEA/D-RDG [21]. Besides, differential grouping used in CCLSM [22], and variable analysis used in MOEA/DVA [23] and DPCCMOEA [24] are another two popular grouping strategies for solving LSMOPs. Note that CCLSM and DPCCMOEA are based on the co-evolutionary framework same as CCGDE.

The second type of MOEAs aims to reduce the decision space via problem transformation or dimensionality reduction. The main idea of problem transformation is to find a proxy task

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by converting an LSMOP into a small-scale problem, where the optimal variant of a given solution can be found in the small-scale problem. The representative MOEAs include LSMOF [25] and LMOEA-DS [26]. Differently, dimensionality reduction-based MOEAs utilize different reduction techniques to reduce the number of variables, such as random embedding in ReMO [27], unsupervised learning in MOEA/PSL [28], and pattern mining in PM-MOEA [29].

The third type of MOEAs focuses on solving LSMOPs by designing various novel search strategies. Some MOEAs are devoted to devising different novel operators to improve the ability to search, the representatives include SparseEA [30] introducing sparse genetic operators and DGEA [31] based on adaptive offspring generation. Other MOEAs are designed to hold a probability model, where the offspring individuals are generated via the devised probability model. The representatives contain IM-MOEA [32] and GMOEA [33], which are based on Gaussian process-based inverse model and generative adversarial networks, respectively.

In summary, the existing large-scale MOEAs have exhibited high effectiveness in solving common LSMOPs [23], [25], sparse LSMOP [28], [30], constrained LSMOP [34], and so on [35], [36], [37], [38]. In addition, there are also many large-scale MOEAs proposed to develop common recommendation systems, such as PMOEA [39], NNIA-RS [40], and MOEA-Probs [41]. Nevertheless, most of these MOEAs mainly focus on how to build a reasonable and effective MOP for their recommendation tasks, and their formulated MOPs are still solved under classic evolutionary frameworks proposed 15 years ago (e.g., NSGA-II [42] and MOEA/D [43]), which cannot effectively solve sparse LSMOPs or constrained LSMOPs as demonstrated in [28], [30], [34].

In this paper, our formulated CMOP for the PEGA task is actually a constrained sparse LSMOP due to the task's intrinsic characteristics. However, there is no MOEA designed for such kind of problem according to [8]. To this end, we design a co-evolutionary framework containing two populations with two encoding from two considerations, where the main population generates offspring individuals by a genetic operator under the guidance of the auxiliary population. On the one hand, as demonstrated in [34], the cooperation of two populations can quickly and accurately find good feasible regions to tackle the difficulty brought by the constraints, since the auxiliary population holds fewer constraints and thus faster convergence speed. On the other hand, the proposed guidance-based genetic operator generates one individual by only flipping several bits of a given parent individual, where the flip operator is effective in solving sparse LSMOPs as demonstrated in [8], [28], [29], [30].

III. Mathematical Formulation of PEGA

A. Problem Statement

Suppose there are N students, M exercises, K knowledge concepts provided by an online education system, which can be represented as $\mathcal{S} = \{s_1, s_2, ..., s_N\}$, $\mathcal{E} = \{e_1, e_2, ..., e_M\}$, and $\mathcal{C} = \{c_1, c_2, ..., c_K\}$, respectively. In addition, there is an exercise-concept relation matrix $Q = (Q_{ij} \in \{0, 1\})^{M \times K}$ measuring the relationship between exercises and knowledge concepts, where $Q_{ij} = 1$ means the exercise e_i is related to knowledge concept c_j, and $Q_{ij} = 0$ otherwise. The $\mathcal{R}$ is used to represent the students' exercise response logs and it can be denoted by a set of triple (s_i, e_j, r_{ij}), where r_{ij} is the response score of student s_i got on exercise e_j, $s_i \in \mathcal{S}$, $e_j \in \mathcal{E}$, and $r_{ij} \in \{0, 1\}$. Here $r_{ij} = 1$ indicates student s_i answered correctly on exercise e_j and $r_{ij} = 0$ otherwise.

The proposed PEGA aims to select m exercises from the exercise bank $\mathcal{E}$ as a group to each student, then improve each student's knowledge proficiency on all knowledge concepts based on the recommended m exercises. The recommended exercise group consisting of m exercises for student s_i can be denoted by

$$U_i = \left\{ u_i^{[1]}, u_i^{[2]}, \ldots, u_i^{[j]}, \ldots, u_i^{[m]} \right\}, \ 1 \le m \le M, \quad (1)$$

where $u_i^{[j]} \in \mathcal{E}$ and it can be also represented by the following binary vector as

$$\mathbf{X}_i = (x_i^1, \ldots, x_i^j, \ldots, x_i^M) \in \{0, 1\}^M, \quad (2)$$

where $x_i^j = 1$ means the exercise e_j is selected and $x_i^j = 0$ indicates the exercise e_j is not selected.

In addition, there is an important vector $\mathbf{n}_i$ for recording the numbers of exercises in U_i (i.e., $\mathbf{X}_i$) related to knowledge concepts, which can be defined as

$$\mathbf{n}_i = (n_i^1, n_i^2, \ldots, n_i^k, \ldots, n_i^K) \quad (3)$$

where each bit n_i^k ranges from 0 to m and indicates the number of exercises in U_i related to knowledge concept c_k, and $\mathbf{n}_i$ can be computed by

$$\begin{aligned} T &= (T_{jk} \in \{0, 1\})^{\sum \mathbf{X}_i \times K} \\ &= (\mathbf{q}_j \mid \forall x_i^j \in \mathbf{X}_i, \text{ if } \mathrm{x_i^j} = 1)^{\sum \mathbf{X}_i \times 1} \\ \mathbf{n}_i &= \left(n_i^k | n_i^k = \sum_{j=1}^{\sum \mathbf{X}_i} T_{jk} \right)^{1 \times K}, \end{aligned} \quad (4)$$

where $\sum \mathbf{X}_i$ is the sum of all items in $\mathbf{X}_i$ denoting the number of selected exercises, i.e., $\sum \mathbf{X}_i = m$, and $\mathbf{q}_j = (Q_{j1}, Q_{j2}, \ldots, Q_{jK})$ is a concept vector related to exercise e_j. The first equation is used to extract the rows of matrix Q corresponding to exercises included in $\mathbf{X}_i$, while the second equation is actually the row sum operation of T to calculate each n_i^k.

B. Formulated Multi-Objective Optimization Problem

In fact, a good exercise group recommendation for a student should enhance the student's proficiency as high as possible after the student practices the recommended exercises. However, it is impractical to measure the quality of the recommended exercise group by making a real student practice the recommended exercise, and there is also no open-available method to measure the quality of the recommended exercise group.

To start with, we have a simple but intuitive assumption that the improvement extent of the student's proficiency on each

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knowledge concept is highly correlated with the number of recommended exercises on each knowledge concept. Therefore, to make the recommended exercise group U_i can help student s_i enhance student's proficiency on knowledge concepts, especially on these knowledge concepts that the student masters poorly, the distribution of exercises in U_i over knowledge concepts should be highly correlated to that of the student s_i's proficiency on the knowledge concepts. Specifically, the poorer a knowledge concept the student masters, the more exercises containing this knowledge concept should be recommended, and fewer exercises should be recommended if some knowledge concepts have been well mastered by student s_i.

For this aim, we employ *Pearson Correlation Coefficient* [44] to measure the above correlation as follows:

$$O_1(\mathbf{n}_i, \boldsymbol{\alpha}_i) = \frac{1}{(K-1)} \cdot \sum_{k=1}^{K} \frac{\left(\alpha_i^k - \bar{\boldsymbol{\alpha}}_i\right)}{\sigma \boldsymbol{\alpha}_i} \cdot \frac{\left(n_i^k - \bar{\mathbf{n}}_i\right)}{\sigma \mathbf{n}_i}, \quad (5)$$

where $\alpha_i \in R^{1\times K}$ is a real vector with a length of K representing the knowledge proficiency of student s_i. $\bar{\boldsymbol{\alpha}}_i$ and $\sigma\boldsymbol{\alpha}_i$ are the mean and standard deviation of $\boldsymbol{\alpha}_i$, respectively, and it is the same case for $\mathbf{n}_i$. The smaller value of $O_1(\mathbf{n}_i, \boldsymbol{\alpha}_i)$ means the higher correlation between $\mathbf{n}_i$ and $\boldsymbol{\alpha}_i$, which further indicates that the recommended exercises in U_i contain more knowledge concepts the student masters poorly. Thus we take $O_1(\mathbf{n}_i, \boldsymbol{\alpha}_i)$ as our **weakness consolidation** objective (O_1).

In addition to consolidating student's weakness, the recommended exercise group U_i should also provide some exercises, which contain new knowledge concepts that the student never saw as many as possible. To this end, we propose the second objective: **new knowledge exploration** (O_2), which can be represented by

$$O_2(\mathbf{n}_i) = \frac{\sum(\mathbf{n}_i \neq 0) - \sum(\,(\mathbf{n}_i \neq 0) \odot \boldsymbol{\mu}_i)}{K - \sum \boldsymbol{\mu}_i + \epsilon}, \quad (6)$$

where $\odot$ is the element-wise product and ϵ is a very small number. $(\mathbf{n}_i \neq 0) \in \{0,1\}^{1\times K}$ is based on a logical operation and outputs a binary vector, which indicates whether knowledge concepts are contained in U_i. $\boldsymbol{\mu}_i \in \{0,1\}^{1\times K}$ represents whether student s_i has done the corresponding knowledge concepts. In theory, $\sum(\mathbf{n}_i \neq 0)$ ranges from 1 to K and $\sum((\mathbf{n}_i \neq 0) \odot \boldsymbol{\mu}_i)$ ranges from 0 to K. $\sum \boldsymbol{\mu}_i$ ranges from 0 to K and denotes the sum of all items in $\boldsymbol{\mu}_i$, where the sum operation has same effect on other vectors. The bigger value of $O_2(\mathbf{n}_i)$ means the more new knowledge concepts are included in U_i, which can promote the student to explore new knowledge concepts.

For further improving the diversity of knowledge concepts included in U_i, a **knowledge diversity** objective (O_3) is proposed, whose formal definition is as follow:

$$O_3(\mathbf{n}_i) = \frac{\sum(\mathbf{n}_i \geq 1)}{K}, \quad (7)$$

where $\sum(\mathbf{n}_i \geq 1)$ ranges from 1 to K and a bigger value of $O_3(\mathbf{n}_i)$ means a better diversity of contained knowledge concepts.

Equipped with the above three objectives, in this paper we formulate the PEGA task as a multi-objective optimization problem as follows:

$$\min_{\mathbf{X}_i} F(\mathbf{X}_i, s_i) = \begin{cases} f_1(\mathbf{X}_i, s_i) = O_1(\mathbf{n}_i, \boldsymbol{\alpha}_i) \\ f_2(\mathbf{X}_i) = -O_2(\mathbf{n}_i) \\ f_3(\mathbf{X}_i) = -O_3(\mathbf{n}_i) \end{cases}, \boldsymbol{\alpha}_i = CD(s_i), \quad (8)$$

where $\mathbf{n}_i$ is computed from $\mathbf{X}_i$ based on (4), $\boldsymbol{\alpha}_i = CD(s_i)$ denotes getting student s_i's knowledge proficiency $\boldsymbol{\alpha}_i$ via a CD model $CD(\cdot)$.

In addition, due to the intrinsic characteristics of the PEGA task, there are four constraints to the formulated problem in (8), which are as follows:

$$\begin{aligned} &\min_{\mathbf{X}_i} F(\mathbf{X}_i, s_i) = (f_1(\mathbf{X}_i, s_i), f_2(\mathbf{X}_i), f_3(\mathbf{X}_i)) \\ &\text{subject to} \quad C_1: \sum \mathbf{X}_i = m \\ &\qquad\qquad\quad C_2: \sum(\mathbf{X}_i \odot \boldsymbol{\nu}_i) = 0 \\ &\qquad\qquad\quad C_3: l \leq y_i^j \leq h, \text{for all } x_i^j = 1 \\ &\qquad\qquad\quad C_4: L \leq \left(\frac{\sum_{j=1}^{M} x_i^j * y_i^j}{m}\right) \leq H, \end{aligned} \quad (9)$$

where $\boldsymbol{\nu}_i \in \{0,1\}^{1\times M}$ indicates whether the student s_i has done the corresponding exercise, y_i^j is the probability of student s_i answering correctly on exercise e_j, and $\sum_{j=1}^{M} x_i^j * y_i^j \in [0,1]$. The first constraint C_1 ensures that the number of selected exercises in U_i is equal to m, the second constraint C_2 enables the exercises that have been done by student s_i not to be selected, the third constraint C_3 makes each selected exercise not be too easy or too difficult, and the final constraint C_4 is used to guarantee the overall correct rate of student s_i on the recommended exercise group U_i to be a high level. Note that the first two constraints are basic constrained conditions caused by the innate character of this problem, and the other two constraints aim to make the student maintain a high interest in learning.

To solve the CMOP presented in (9) for each student, there are two key challenges that need to be overcome. The first is to diagnose the accurate knowledge proficiency α_i for the student based on a CD model, which is necessary for computing the $f_1(\cdot)$ value in (8), but existing CD models cannot provide the diagnosed knowledge proficiency on the knowledge concepts that are not correlated to the student's response records. Since the proposed CMOP is actually a constrained sparse LSMOP where M is large and $m \ll M$, and existing MOEAs cannot solve such constrained sparse LSMOP well. Therefore, the second challenge is to design an effective MOEA to solve the proposed CMOP. In the following subsection, we will present the extended neural CD model for the first challenge, and we will introduce a co-evolutionary algorithm to overcome the second challenge in Section IV.

C. Proposed Extended Neural CD

Due to the sparse interactions between a student and knowledge concepts, existing CD approaches cannot diagnose the student's proficiency on all knowledge concepts, and the diagnosed proficiency will lead to the inaccurate computation of (8). To

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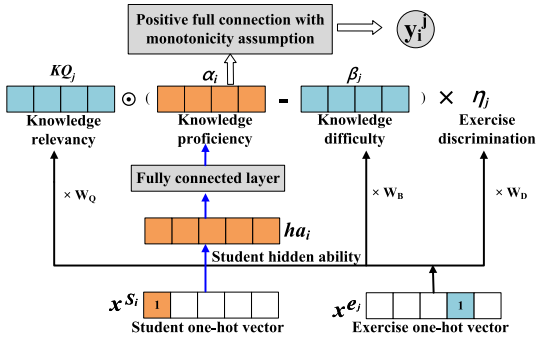


Fig. 3. The proposed extended neural CD model (ENCD). To model student's proficiency on all knowledge concepts, an extra fully connected layer is applied to student's hidden ability vector $\mathbf{ha}_i$.

address this issue, we design a novel CD model by extending the NeuralCD [6], which is named extended neural CD (termed ENCD).

Instead of directly modeling $\boldsymbol{\alpha}_i$ via a simple embedding layer that multiplies the student's one-hot vector $\mathbf{x}^{s_i} \in \{0,1\}^{1\times N}$ with a trainable matrix, as shown in Fig. 3, an extra fully connected (FC) layer is added to capture the complex latent representation for the student's knowledge proficiency, which can be defined as follows:

$$\begin{aligned} \mathbf{ha}_i &= FC_1(\mathbf{x}^{s_i}),\ \mathbf{ha}_i \in R^{1\times d} \\ \boldsymbol{\alpha}_i &= FC_2(\mathbf{ha}_i),\ \boldsymbol{\alpha}_i \in R^{1\times K}, \end{aligned} \tag{10}$$

where d is the embedding size, $\mathbf{ha}_i$ is the student's hidden ability representation vector, and $\boldsymbol{\alpha}_i$ is the student's knowledge proficiency on all knowledge concepts.

Except for the computation of student's knowledge proficiency, the other parts of the proposed ENCD keep the same as that of NeuralCD. To be specific, the knowledge relevancy KQ_j, knowledge difficult β_j, and exercise discrimination η_j can be obtained by

$$\begin{aligned} KQ_j &= \sigma(\mathbf{x}^{e_j} \times \mathbf{W_Q}), \mathbf{W_Q} \in R^{M\times d} \\ \beta_j &= \sigma(\mathbf{x}^{e_j} \times \mathbf{W_B}), \mathbf{W_B} \in R^{M\times d} \\ \eta_j &= \sigma(\mathbf{x}^{e_j} \times \mathbf{W_D}), \mathbf{W_D} \in R^{M\times 1}, \end{aligned} \tag{11}$$

where $\mathbf{W_Q}$, $\mathbf{W_B}$, and $\mathbf{W_D}$ are trainable matrices, $\mathbf{x}^{e_j} \in \{0,1\}^{1\times M}$ denotes the one-hot vector for exercise e_j, and $\sigma(\cdot)$ is the Sigmoid activation function.

With the above obtained elements, the probability of student s_i answering correctly on exercise e_j is computed by the final diagnosis layer as

$$\begin{aligned} temp &= KQ_j \odot (\boldsymbol{\alpha}_i - \beta_j) \times \eta_j \\ y_i^j &= FC_5(FC_4(FC_3(temp))), \end{aligned} \tag{12}$$

where the first equation is used to get the aggregation results, and three FC layers are used to get the final output y_i^j.

To train the proposed ENCD, the Cross-Entropy loss function is adopted:

$$\mathcal{L} = -\sum_i \sum_j \left(r_{ij} \log y_i^j + (1 - r_{ij}) \log \left(1 - y_i^j\right)\right). \tag{13}$$

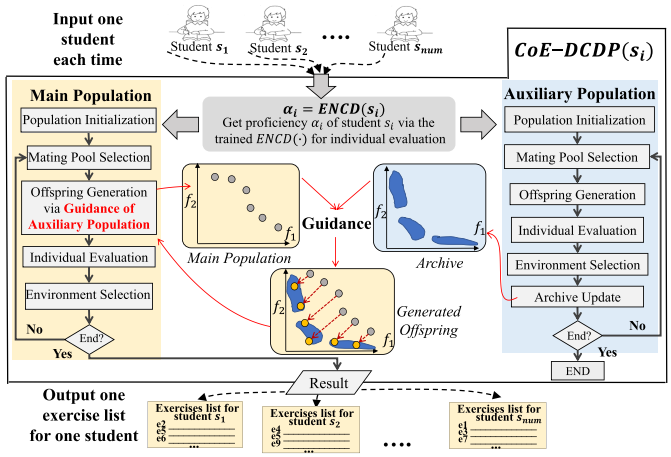


Fig. 4. The overall procedure of the proposed CoE-DCDP for each students, where the $ENCD(\cdot)$ is trained based on all students' response logs and the proposed CoE-DCDP receives one student as its input each time. The offspring individuals in the main population will be generated under the guidance of archive individuals collected from the auxiliary population. After the evolution, one of the individuals in the main population will be selected and decoded into the exercise group, which will be recommended to the student.

Same as NeuralCD, the training process is completed with a standard optimizer. After training, the value of $\boldsymbol{\alpha}_i$ is the diagnosis result for each student, which denotes the student's proficiency on all knowledge concepts.

IV. The Proposed CoE-DCDP

A. Overall Framework of CoE-DCDP

Based on the trained ENCD model, the main idea of the proposed CoE-DCDP is to solve the problem in (9) by the cooperation of two populations with two kinds of encoding. One population equipped with exercise binary encoding, i.e., the main population, is responsible for fine search (known as exploitation), since the exercise group changes slightly when a small change happens in this encoding. Another population equipped with knowledge concept integer encoding, i.e., the auxiliary population, is employed for rough search (known as exploration), since a small change in this encoding can cause a great change in the exercise group, and much fewer constraints need to be solved when using this encoding. Since the auxiliary population can quickly find some potential areas in the early stage of evolution, the convergence of the main population can be accelerated by the guidance of the auxiliary population, where a novel flip operator is devised to solve the sparsity difficulty of the CMOP.

For better understanding, Fig. 4 has summarized the overall framework of the proposed CoE-DCDP, which receives one student each time as the input and outputs one group of exercises (i.e., exercises list) for the student. First, the student's knowledge proficiency α_i is diagnosed via the trained ENCD, where the obtained α_i is used for individual evaluation in two populations. Second, the main population and the auxiliary population start to evolve, where two populations are initialized by different initialization strategies. Third, a binary tournament selection is

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Algorithm 1: Main Steps of CoE-DCDP.

Input: Gen: Maximum number of generations; Pop: Population size; s_i: A student's one-hot vector;
Output: $\mathbf{P}_m^i$: One individual in the main population;

$\alpha_i \leftarrow ENCD(s_i)$; % Get student's proficiency via trained ENCD
$\mathbf{P}_m, \mathbf{P}_a, A \leftarrow$ Initialization$_1(Pop)$, Initialization$_2(Pop)$, $\emptyset$; % Initialize two populations and set the archive
for $g = 1$ to Gen **do**
  Update $Ratio$ via a parameter adaptation strategy;
  $\mathbf{P}_m{}', \mathbf{P}'_a \leftarrow$ Select Pop parents from $\mathbf{P}_m$ and $\mathbf{P}_a$, respectively; % Mating pool selection
  ————**Auxiliary population**————
  $\mathbf{Off}_a \leftarrow$ Genetic Operator$(\mathbf{P}'_a)$;
  Evaluate individuals in $\mathbf{Off}_a$ according to (8) and α_i; $\mathbf{P}_a \leftarrow$ EnvironmentSelection$(\mathbf{P}_a \cup \mathbf{Off}_a)$;
  $A \leftarrow$ ArchiveUpdate$(A \cup \mathbf{P}_a)$;
  ————**Main population**————
  $\mathbf{Off}_m \leftarrow$ Guidance-based Genetic Operator$(\mathbf{P}_m{}', A, Ratio, Pop)$; % Algorithm IV-C
  Evaluate individuals in $\mathbf{Off}_M$ according to (8) and α_i;
  $\mathbf{P}_m \leftarrow$ EnvironmentSelection$(\mathbf{P}_m \cup \mathbf{Off}_m)$;
end for
Select one individual $\mathbf{P}_m^i$ from $\mathbf{P}_m$ as the output
return $\mathbf{P}_m^i$;

employed to mate individuals for each population. Next, the auxiliary population will execute offspring generation, individual evaluation, environment selection, and archive update, while the main population is blocked to wait for the archive updated by the auxiliary population, where the Pareto-optimal individuals of the auxiliary population are kept. Fifth, the mated parent individuals in the main population generate offspring individuals based on the guidance of the individuals in the archive, and each offspring individual needs to be evaluated. Finally, the environment selection will be applied to two populations to maintain potentially good individuals. The second to the sixth step will be repeated until the evolution termination criterion is satisfied, after which one individual will be output as the exercise list for the input student. For details, Algorithm 1 demonstrates main steps of the proposed CoE-DCDP.

B. Encoding Strategy

There are two types of encoding respectively used for two populations, including the exercise binary encoding and the knowledge concept integer encoding. The exercise binary encoding employs a binary vector $\mathbf{X}_i$ in (2), while an integer vector $\mathbf{n}_i$ in (3) is used for the knowledge concept integer encoding.

For better observation, Fig. 5 presents an illustrative example of two types of encoding. As can be seen, the recommended $m = 5$ exercises $U_i = \{e_1, e_2, e_6, e_8.e_{10}\}$ can be represented by

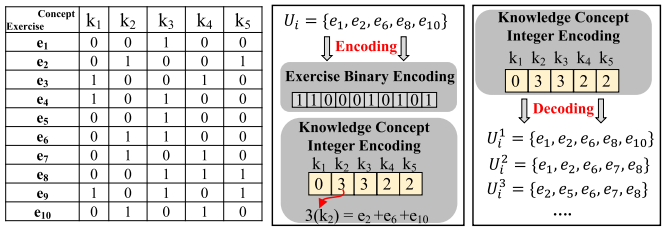


Fig. 5. An illustrative example of proposed encoding strategy. Herein, the number of exercises in U_i is set to 5 (m=5). In general, multiple different exercise sets can be represented by the same knowledge concept integer encoding.

its corresponding exercise binary encoding (1,1,0,0,0,1,0,1,0,1) and knowledge concept integer encoding (0,3,3,2,2), where the second bit of integer encoding equal to 3 means that there are 3 exercises related to the second concept k_2 including e_2, e_6, and e_{10}. Different from the binary encoding, one knowledge concept integer encoding could be decoded into multiple different exercise sets. For example, for (0,3,3,2,2), the second bit equal to 3 means that 3 exercises related to concept k_2 should be included in one subset, which could be $\{e_2, e_6, e_{10}\}$ or $\{e_2, e_6, e_7\}$ or others. Similarly, other bits may also correspond to multiple different exercise subsets. Finally, the feasible decoded exercise set could be U_i^1 or U_i^2 or U_i^3 or others, which is built by combining these obtained exercise subsets.

As can be seen, the individual under knowledge concept integer encoding can correspond to multiple exercise sets. When a small change happens in the individual represented by the exercise binary encoding, the corresponding exercise set only changes slightly, while the related exercise sets will change greatly when there is a small change happened in the individual represented by the knowledge concept integer encoding. Therefore, the knowledge concept integer encoding will enable the population easily to get out of local optima and explore more new areas, which is known as the ability of exploration [45]. Due to the different natures of the two encoding, in the proposed CoE-DCDP, the auxiliary population is equipped with the knowledge concept integer encoding to perform a rough search for exploration, which aims to search the optimal distribution of exercises on knowledge concepts and thus provides fast convergence, while the main population is equipped with the exercise binary encoding to perform a fine search for exploitation, which aims to determine which exercises are selected as the recommendation and thus provides good diversity of exercise sets.

It is worth noting that the CMOP in (9) will degenerate into the following problem for the auxiliary population:

$$\min_{\mathbf{n}_i} F(\mathbf{n}_i, s_i) = (f_1(\mathbf{n}_i, s_i), f_2(\mathbf{n}_i), f_3(\mathbf{n}_i))$$
$$\text{subject to } C_1 : \sum \mathbf{n}_i = m. \tag{14}$$

Here, only one constraint is available due to the unique properties of the integer encoding, which also makes the auxiliary population hold a faster convergence speed.

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C. Guidance-Based Offspring Generation

As mentioned above, the auxiliary population using the knowledge concept integer encoding holds a faster convergence speed but worse diversity of exercise sets than the main population. To make the best of two populations, we propose a guidance-based genetic operator for the main population to accelerate its convergence as well as hold high diversity. Specifically, individuals in archive A of the auxiliary population will be employed to guide the crossover and mutation of individuals in the main population, where each offspring individual needs to make its distribution of exercises over knowledge concepts be consistent with the corresponding guidance individual from A through a novel flip operator.

Given an offspring individual (denoted by I_{off}) and a guidance individual (denoted by $I_{guidance}$), their knowledge concept integer encoding is Int-En(I_{off}) and Int-En($I_{guidance}$), where Int-En(I_{off}) is obtained based on (4). Then the difference in the number of exercises over knowledge concepts between I_{off} and $I_{guidance}$ can be computed as follows:

$$\begin{aligned}\mathbf{Diff} &= \text{Int-En}(I_{guidance}) - \text{Int-En}(I_{off}) \\ &= (Diff_1, Diff_2, \ldots, Diff_K),\end{aligned} \tag{15}$$

where $Diff_1$ represents the difference in the number of exercises over knowledge concept k_1 between I_{off} and $I_{guidance}$, and further means there should be $Diff_1$ more exercises containing k_1 that need to be included in I_{off}.

Suppose there are $N_{kj} = \sum_{i=1}^{M} Q_{ij}$ exercises related to knowledge concept k_j in exercise set $\mathcal{E}$, and I_{off} selects N_{kj}^{off} exercises on knowledge concept k_j, then we propose the following two probabilities of flipping as

$$Pof_{kj}^1 = \frac{p_c * N_{kj} + Diff_j}{2 * N_{kj}^{off}},\ Pof_{kj}^0 = \frac{p_c * N_{kj} - Diff_j}{2 * (N_{kj} - N_{kj}^{off})}, \tag{16}$$

where p_c is the crossover probability commonly set to 0.5, Pof_{kj}^1 is the probability of flipping zeros related to k_j in I_{off} to ones, and Pof_{kj}^0 refers to the probability of flipping ones related to k_j in I_{off} to zeros. As can be observed from Pof_{kj}^1, the big value of $Diff_j$ and small value of N_{kj}^{off} will lead to more zeros flipped to ones, which means that more exercises on concept k_j will be selected in I_{off}, and it is the same case for Pof_{kj}^0.

Next, two random vectors $R0 \in [0,1]^{1\times(N_{kj}-N_{kj}^{off})}$ and $R1 \in [0,1]^{1\times N_{kj}^{off}}$ are generated to determine which zeros or ones related to k_j need to be flipped by

$$Ff_{kj}^1 = R0 < Pof_{kj}^1,\ Ff_{kj}^0 = R1 < Pof_{kj}^0, \tag{17}$$

where $Ff_{kj}^1 \in \{0,1\}^{1\times(N_{kj}-N_{kj}^{off})}$ and $Ff_{kj}^0 \in \{0,1\}^{1\times N_{kj}^{off}}$ are two flipping flags, and the flag value equal to 1 means executing the flipping operation.

Based on the above-tailored flipping operation, Algorithm IV-C lists the detailed procedure of the proposed guidance-based genetic operator. First, two individuals I_{off} and I_{off1} are selected from mated parent population (Line 3). Then, a randomly generated number $rand$ is compared to $Ratio$ which is the ratio of offspring individuals generated with the guidance of A (Line 4). If $rand > Ratio$, the widely used single-point crossover and bit-wise mutation [30] will be applied to I_{off} and I_{off1}, where a new offspring individual (also denoted by I_{off}) will be generated (Line 5), otherwise a guidance individual $I_{guidance}$ will be randomly selected from archive A (Line 7). Third, the **Diff** between I_{off} and $I_{guidance}$ is computed based on (15) (Line 9). Next, the flipping operation in (17) is executed to I_{off} according to two probabilities Pof_{kj}^1 and Pof_{kj}^0 obtained from (16) (Lines 10-11), and the unflipped bits related to k_j in I_{off} will be randomly flipped to keep consistent with $I_{guidance}$, i.e., $Diff_j = 0$, (Line 12). Note that the third to the fourth step will repeat for all knowledge concepts. Fifth, the generated I_{off} (from the second step or fourth step) has to repair its constraints by a local search technique, which will be presented in Section IV-E3. Finally, the adjusted I_{off} will be added to offspring population Off.

Algorithm 2: Guidance-Based Genetic Operator().

Input: $\mathbf{P}'_m$:Mated parent population, A: Archive, $Ratio$: Ratio of offspring generated with the guidance of A; Pop: Population size
Output: Off: Offspring;

1: $Off \leftarrow \emptyset$;
2: **for** $i = 1$ to Pop **do**
3: $I_{off}, I_{off1} \leftarrow \mathbf{P}'_m[i,:], \mathbf{P}'_m[i+1,:]$;
4: **if** rand $> Ratio$ **then**
5: $I_{off} \leftarrow$ Apply single-point crossover and bitwise mutation to I_{off} and I_{off1};
6: **else**
7: $I_{guidance} \leftarrow A$[random,:]; % `Random selection`
8: **for** $j = 1$ to K **do**
9: **Diff** $\leftarrow$Compute the difference via (15);
10: Pof_{kj}^1, $Pof_{kj}^0 \leftarrow$ Get two probabilities by (16);
————**Crossover**————
11: Flip zeros and ones according to (17);
————**Mutation**————
12: Randomly flip rest bits to make $Diff_j$ to be zero;
13: **end for**
14: **end if**
15: $I_{off} \leftarrow$Repair constraints of I_{off} by local search;
16: $Off \leftarrow Off \cup I_{off}$;
17: **end for**
18: **return** Off;

For better understanding, Fig. 6 gives an illustrative example of the guidance-based genetic operator. Given an individual $I_{off} = (1,0,1,0,0,0,1,0,1,0)$ and a guidance individual $I_{guidance} = (1,3,1)$, the knowledge concept integer encoding of I_{off} is obtained as Int-En(I_{off}) $= (2,1,2)$. The concept k_1 is related to 4 exercises: e_3, e_4, e_5, and e_9, where corresponding decision variables (1,0,0,1) are extracted. The next is to compute the two probabilities of flipping $Pof_{k1}^1 = \frac{1}{4}$ and $Pof_{k1}^0 = \frac{3}{4}$, randomly generate two vectors $R0 = (0.2, 0.6)$ and

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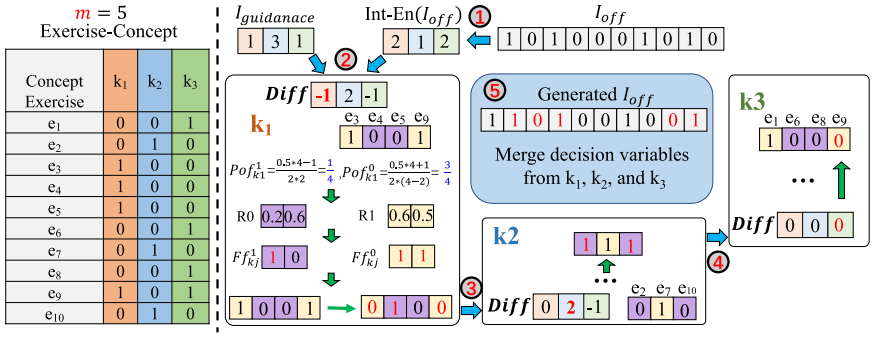


Fig. 6. An illustrative example of guidance-based genetic operator. For the decision variables in the individual I_{off} related to one knowledge concept, they have to change to make the exercise distribution over this knowledge concept in I_{off} be same as that in guidance individual $I_{guidance}$. After executing the above operation for all knowledge concepts, the individual I_{off} will turn to be an offspring individual.

$R1 = (0.6, 0.5)$, and then obtain two flipping flags $Ff^1_{k1} = (1, 0)$ and $Ff^0_{k1} = (1, 1)$. With the two flags, the corresponding bits of extracted variables (1,0,0,1) are flipped, where new variables (0,1,0,0) are formed. The above procedure is repeated for concept k_2 and k_3, where all newly formed variables (0,1,0,0), (1,1,1), and (1,0,0,0) will be merged as a new individual (1,1,0,1,0,0,1,0,0,1). Note that the bits corresponding to e_2, e_3, e_4, e_9, and e_{10} are flipped.

D. Recommendation Strategy

Due to multiple non-dominated individuals (solutions) found by the MOEA in one run, it is necessary to select one suitable solution for final recommendation. For example, the solutions located in the knee point area [46] are usually preferred since they strike a balance among different objectives. However, in this paper, the solution with the best value of weaknesses consolidation will be finally recommended due to the task request.

E. Other Techniques

In the proposed CoE-DCDP, the mating pool selection and the environment selection are same as that of SPEA2 [47], which demonstrates pretty good effectiveness on the MOPs. The single-point crossover and bit-wise mutation are used as the genetic operator in the auxiliary population. In addition, there are other three techniques, including initialization strategy for the auxiliary population, parameter adaptation for $Ratio$, and constraints repair technique.

1) Initialization Strategy: For better initial diversity and convergence of the auxiliary population, more exercises for knowledge concepts that the student masters poorly (i.e., weak concepts) should be selected. To achieve this, each individual has to compute the average number of exercises on weak concepts as

$$AvgNum_{weak} = \text{round}\left(\frac{m * U(\cdot)}{K/2}\right), \tag{18}$$

where $U(\cdot)$ is a uniform distribution between 0.5 and 1, and $\text{round}(\cdot)$ is the round operation. Then the average number of exercises on strong concepts can be also inferred. All individuals in the auxiliary population are initialized based on the above idea.

2) Parameter Adaptation for Ratio: To achieve a good balance between exploration and exploitation, a parameter adaptation strategy for $Ratio$ is designed as

$$Ratio = 0.5 * \left(1 - \cos\left(\left(1 - \frac{g}{Gen}\right) * \pi\right)\right), \tag{19}$$

where g is the current generation and Gen is the maximal number of generations. By doing so, the main population depends more on the guidance of the auxiliary population in the early evolution stage for fast convergence, while in the late stage the main population will gradually take the advantage of the binary encoding to get out of local optima for good diversity and convergence.

3) Local Search-Based Constraint Repair: Due to the simple rules that can deal with constraints C_2 and C_3, there are merely two sub-techniques for repairing constraints C_1 and C_4.

Repair C_1: If the number of selected exercises is more than m, randomly flip extra ones to zeros, otherwise, randomly flip some zeros to ones to satisfy the constraint.

Repair C_4: If the overall correct rate on selected exercises is lower than L, more exercises that are easy to answer will be used to replace these selected exercises that are difficult to answer, otherwise, more difficult exercises are used to replace these selected easy exercises. The difficulty degree of an exercise is judged according to the probability predicted by the proposed ENCD.

F. Time Complexity Analysis

The time complexity of the proposed CoE-DCDP is mainly determined by three components, i.e., the calculation of three objectives, the genetic operator used for each population, and the remained optimization process of SPEA2. Based on previously defined notations, the time complexity of O_1, O_2, and O_3 is $O(K)$ [44], $O(1)$, and $O(1)$, totally $O(Pop \times (K + 2))$ for one generation. The time complexity of genetic operator in one generation is $O(Pop \times M)$, and the time complexity of one generation of SPEA2 is $O(4 \times Pop^2)$[47]. Therefore, the overall time complexity of CoE-DCDP is $O(2 \times Gen \times Pop \times (K + 2 + 4 \times Pop + M))$. Since $M \gg (K + 2 + 4 \times Pop)$ in general, the time complexity of CoE-DCDP can be regarded as $O(2 \times Gen \times Pop \times M)$. Compared to the recently published

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TABLE I
DATASET STATISTICS

Dataset	# Students	# Concepts	# Exercises	# Response logs
ASSISTments	4,163	123	17,746	324,572
JunYi	1,000	39	712	202,945
SLP	1,499	34	907	57,244

recommendation algorithm SGL [48], the proposed CoE-DCDP holds better computational efficiency since the time complexity of SGL $O(M \times m \times (M+m))$ is much higher than that of CoE-DCDP.

V. EXPERIMENTAL STUDIES

A. Experimental Settings

1) Benchmark Datasets: Three widely used datasets ASSISTments[2], JunYi [49], and SLP[3] are used to verify the effectiveness of the proposed CoE-DCDP, whose statistics are summarized in Table I and descriptions are presented as follows:

- ASSISTments is an openly available dataset created in 2004 by the ASSISTments online tutoring service system.
- JunYi is an openly available dataset gathered from JunYi Academy, which is a Taiwan's platform built in 2012.
- SLP is an openly available dataset recently published in 2021, which collected academic performance data of learners during their three-year study on eight different subjects, including Chinese, mathematics, and so on.

2) Performance Metrics: As introduced in Section III-B, there is no open-available measure method to test the score of the recommended exercise group for a student. Based on the assumption that the improvement extent of the student's proficiency on each knowledge concept is highly correlated with the number of recommended exercises on each knowledge concept, the quality of the recommended exercise group is determined by whether more exercises are selected for student's real weak knowledge concepts and fewer exercises are selected for student's real strong knowledge concepts. In other words, the distribution of the recommended exercises over knowledge concepts should be highly correlated to that of the student's proficiency on the knowledge concepts. However, there is no ground truth due to unavailability of the student's real ability/proficiency, which is different from traditional goods recommendation and movie recommendation, thus traditional metrics such as precision and recall cannot be employed.

To address the issue, we execute a simulation experiment to build three response datasets for ASSISTments, JunYi and SLP. Each response dataset is composed of some newly added students and their response logs, where the newly added student's real ability/proficiency is randomly generated in advance. Specifically, the NeuralCD [6] model is first employed to be trained based on response logs in ASSISTments or JunYi or SLP, where there are two types of parameters in NeuralCD: student's ability parameter $Para_a$ and exercise parameter $Para_e$. To generate response logs of a new student s_{new}, then a randomly generated vector is used to replace $Para_a$, which is regarded as s_{new}'s real ability/proficiency. In the following, the NeuralCD with replaced $Para_a$ is used to predict s_{new}'s answer to each exercise e_j of $\mathcal{E}$ in ASSISTments or JunYi or SLP, where the output is greater than 0.5 means the answer is correct ($r_{new\ j} = 1$), otherwise $r_{new\ j} = 0$. Finally, student s_{new}'s response logs can be built as $(s_{new}, e_j, r_{new\ j})$. The second to the fourth step will repeat for Num_{add} times to add Num_{add} students and their response logs to original ASSISTments or JunYi or SLP. By doing so, the students' proficiency generated in advance can be seen as student's real proficiency (i.e., the ground truth), where Num_{add} students have their own ground truth that can be used for algorithm comparison.

Note that students' real proficiency on each knowledge concept is measured by the value of corresponding bit in the student's ability parameter, where a big value indicates good proficiency. The better the student's real proficiency on one knowledge concept is, the fewer exercises related to this knowledge concept should be recommended. Thus, the lower correlation between the student's real proficiency and the distribution of exercises in the recommended exercise group over knowledge concepts is, the better the quality of the recommended exercise group is. To measure such a correlation, three widely adopted metrics are employed, which are as follows:

- *Pearson Correlation Coefficients* [44].
- *Spearman Rank Correlation Coefficients* [50].
- *Kendall'τ Rank Correlation Coefficients* [51].

In addition, there are other metrics that will be briefly introduced in the corresponding sections.

3) Compared Approaches: There are three groups of state-of-the-art comparison approaches for validation, including exercise group recommendation approaches, MOEAs, and CD approaches.

To verify the effectiveness of the proposed CoE-DCDP, state-of-the-art exercise group recommendation approaches are used to compare their recommended exercise groups. However, because there is no baseline approach for the PEGA task, we directly employ two traditional recommendation approaches as the baseline, and other MOEA-based approaches optimizing the same objectives as our approach are also considered as the baseline. The followings are their descriptions:

- CF: CF [52] is a traditional collaborative filtering recommendation algorithm, where a model will output the probability of the student answering correctly each exercise of the exercise set and recommend m exercises with the lowest probability for the student to consolidate the student's weak knowledge concepts.
- SGL: SGL [48] is a recently published recommendation algorithm based on graph self-supervised learning, which like CF also recommend m exercises for the student to consolidate the weakness. Differently, not only the probability of each exercise but also the diversity of contained concepts are considered during the recommendation.

[2]https://sites.google.com/site/assistmentsdata/
[3]https://aic-fe.bnu.edu.cn/en/data/index.html

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TABLE II
THE COMPARISON OF THE PROPOSED COE-DCDP AND COMPARED APPROACHES IN TERMS OF THREE CORRELATION METRICS: PEARSON, SPEARMAN, AND KENDALL ON BOTH ASSISTMENTS, JUNYI AND SLP, AVERAGED ON 100 STUDENTS

Dataset	Metric	CF	SGL	MOEA-Probs -NCD	SparseEA -NCD	CCMO -NCD	CoE-DCDP -NCD	MOEA-Probs -ENCD	SparseEA -ENCD	CCMO -ENCD	CoE-DCDP
ASSISTments	Pearson	-0.0273	-0.0172	-0.0011	-0.0865	-0.0336	-0.1792	-0.1716	-0.2518	-0.1508	-0.5589
	Spearman	0.0186	-0.0496	-0.0118	-0.0793	-0.0448	-0.1698	-0.1628	-0.1949	-0.1187	-0.4771
	Kendall	0.0165	-0.0254	-0.0105	-0.0601	-0.0322	-0.1605	-0.1237	-0.1472	-0.0887	-0.3721
JunYi	Pearson	-0.0581	-0.0837	-0.0491	-0.0201	-0.0084	-0.0981	-0.2235	-0.2121	-0.1486	-0.2396
	Spearman	-0.0241	-0.0628	-0.0649	-0.0175	-0.0411	-0.0116	-0.1975	-0.1803	-0.1242	-0.2132
	Kendall	-0.0446	-0.0601	-0.0744	-0.0312	-0.0450	-0.0325	-0.1801	-0.1637	-0.1091	-0.1867
SLP	Pearson	0.0334	0.0583	0.0419	0.0602	0.0502	0.0605	-0.2847	-0.4267	-0.2456	-0.6017
	Spearman	0.0611	0.0467	0.0571	0.0361	0.0462	0.0524	-0.2342	-0.3553	-0.2312	-0.5421
	Kendall	0.0481	0.0455	0.0435	0.0315	0.0342	0.0452	-0.1864	-0.2416	-0.1817	-0.4193

The lower the correlation is, the better the result is.

- MOEA-Probs-ENCD: MOEA-Probs [41] is a traditional MOEA for recommendation. Herein, we employ the approach to optimize three objectives in (9) that is computed based on ENCD.
- MOEA-Probs-NCD: Same as MOEA-Probs-ENCD, but the objectives in (9) that is computed based on NeuralCD.
- SparseEA-ENCD: SparseEA [30] is an effective MOEA for sparse LSMOPs. Here we directly employ SparseEA to optimize three objectives in (9) that is based on ENCD.
- SparseEA-NCD: Same as SparseEA-ENCD with the objectives computed based on NeuralCD.
- CCMO-ENCD: CCMO [34] is an effective MOEA for constrained LSMOPs under co-evolutionary framework. Here we directly employ CCMO to solve the CMOP in (9) that is based on ENCD.
- CCMO-NCD: Same as CCMO-ENCD with the objectives computed based on NeuralCD.
- CoE-DCDP-NCD: The variant of CoE-DCDP, whose objectives computed based on NeuralCD.

To verify the effectiveness of the proposed co-evolutionary algorithm, state-of-the-art MOEAs are used to compare their searched Pareto fronts, including NSGA-II [42], MOEA/D [43], SparseEA [30], LSMOF [25], CCMO [34], and MOEA/PSL [28].

To verify the effectiveness of the proposed ENCD, state-of-the-art CD approaches are used for comparison, including DINA [4], MIRT [5], and NeuralCD [6].

4) Parameter Settings: Dataset and Model Settings. Only the students whose response records' lengths are more than 15 will be used for experiments, and all remained students in each dataset are randomly split into 70%/10%/20% for training/validation/testing. Num_{add} is set to 100, which leads to 100 randomly generated students added to each dataset. For the model ENCD, d is set to 20 and the dimensions of final three FC layers are set to 512, 256, and 1, respectively.

Algorithm Parameter Settings. The proposed CoE-DCDP will be executed num times for num students, here num is set to 100. The number of exercises m assembled in one exercise group is set to 100 for ASSISTments and set to 50 for JunYi and SLP. Population size Pop is set to 100, the maximal number of generations Gen is set to 400 and 200 for ASSISTments, JunYi and SLP.

For fair comparison, except for the number of evaluations ($Pop \times Gen$), other parameter settings for all compared approaches are same as that in their original papers to hold their best performances.

B. Effectiveness of Exercises Assembled by CoE-DCDP

To demonstrate the effectiveness of the exercises assembled by the proposed CoE-DCDP, Table II compares the proposed CoE-DCDP with compared approaches in terms of three types of correlations between the student's real ability and the exercise distribution in their recommended exercise groups, averaged over 100 students. As can be observed, the correlation coefficients achieved by the proposed CoE-DCDP are significantly lower than that achieved by other approaches, which indicates that our assembled exercises are more suitable for students to enhance their poorly mastered knowledge concepts.

For a deep insight into the results, we take the results of num=100 students on ASSISTments dataset as an illustrate examples in Fig. 7. First, all knowledge concepts are sorted according to the students' real proficiency/ability, where weak knowledge concepts are located on the left of the horizontal axis. Then, we plot the distribution of exercises assembled by the proposed CoE-DCDP and compared approaches, where the vertical axis represents the proportion of recommended exercises for the corresponding knowledge concepts. As can be seen, compared to other nine comparison approaches, the proposed CoE-DCDP has a better distribution of exercises over knowledge concepts, and the proposed CoE-DCDP recommends more exercises for the student's weaker knowledge concepts and fewer exercises for the student's stronger knowledge concepts, which is in line with the goal of strengthening students' weaknesses. Therefore, it can be proved that the proposed approach can achieve the goal of consolidating the weak knowledge concepts of students.

For further investigating the competitiveness of CoE-DCDP, there are two more metrics used for comparison including the novelty and the diversity. Specifically, the novelty metric computed by (6) measures the coverage of new knowledge concepts included in the recommended exercise group, where high novelty coverage rates are usually preferred which means more new knowledge concepts are contained, while the diversity metric

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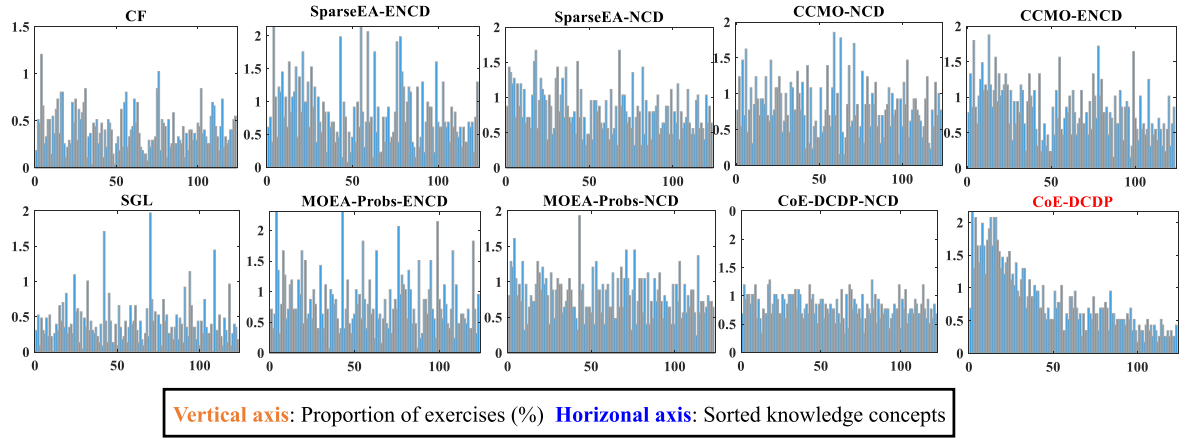
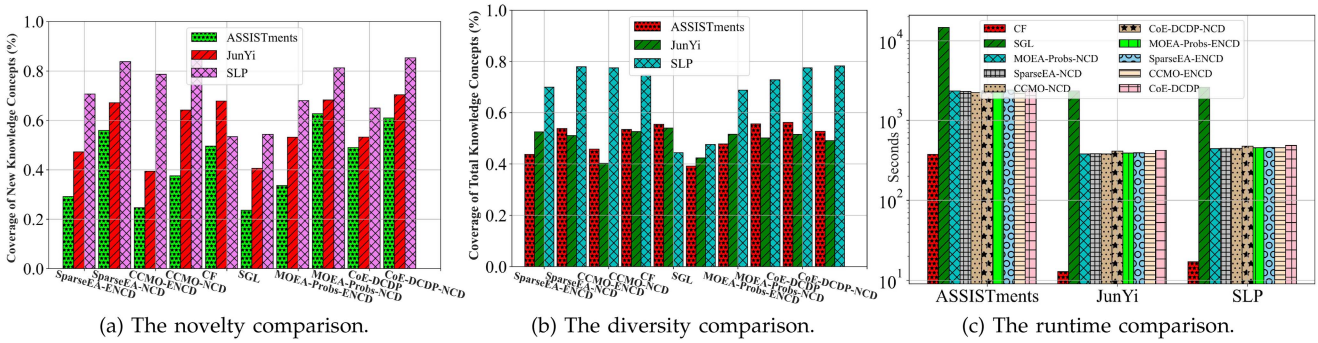


Fig. 7. The comparison of the proposed CoE-DCDP and compared approaches on ASSISTments in terms of the distribution of exercises on all knowledge concepts in num=100 students' recommended exercise groups. Students' weak knowledge concepts are located on the left of the horizontal axis, while their strong knowledge concepts are located on the right part.



(a) The novelty comparison. (b) The diversity comparison. (c) The runtime comparison.

Fig. 8. The novelty, diversity, and runtime comparison between the proposed CoE-DCDP and compared approaches on three datasets.

computed by (7) measures the coverage of total knowledge concepts included in the recommended exercise group, where high diversity coverage rates are usually preferred. Fig. 8(a) and (b) show the comparison of the proposed CoE-DCDP and compared approaches in terms of the novelty and diversity metrics, it can be seen that our approach holds competitively high novelty coverage rates and diversity coverage rates to compared approaches.

To investigate the efficiency of the proposed approach, Fig. 8(c) compares the runtimes of all approaches. It can be found that the runtimes of the proposed CoE-DCDP and other MOEA-based approaches are more than that of CF but significantly less than that of the novel approach SGL. Besides, the proposed approach holds a comparable efficiency to MOEA-based comparison approaches.

In summary, the proposed CoE-DCDP is superior to or competitive with state-of-the-art comparison approaches in terms of efficiency and effectiveness on correlation, novelty, and diversity metrics.

C. Effectiveness of the Proposed ENCD

As can be seen from Table II, all NeuralCD-based approaches including MOEA-Probs-NCD, SparseEA-NCD, CCMO-NCD, and CoE-DCDP-NCD are worse than their own variants, i.e., MOEA-Probs-ENCD, SparseEA-ENCD, CCMO-ENCD, and CoE-DCDP, which are based on the proposed ENCD. The above observation proves the effectiveness of the proposed ENCD to some extent, and the same conclusion can be also obtained from Figs. 7 and 8.

To further investigate the effectiveness of ENCD on cognitive diagnosis, we compare the proposed ENCD with other three state-of-the-art CD approaches, including DINA, MIRT, and NeuralCD. Table III summarizes their testing performances on predicting student's answers to exercises in terms of three metrics [6]: accuracy, RMSE (root mean square error), and AUC (area under curve). As can be seen, the proposed ENCD outperforms DINA, MIRT, and NeuralCD in terms of all indicators on

TABLE III
THE TESTING PERFORMANCE COMPARISON OF THE PROPOSED ENCD AND OTHER CD APPROACHES IN TERMS OF ANSWER PREDICTION

Dataset	ASSISTments/JunYi/SLP		
Model	Accuracy	RMSE	AUC
DINA	0.669/0.676/0.683	0.477/0.461/0.467	0.717/0.752/0.754
MIRT	0.676/0.743/0.739	0.504/0.418/0.448	0.710/0.777/0.771
NeuralCD	0.719/0.748/0.742	0.439/0.408/0.435	0.749/0.797/0.791
ENCD	**0.732/0.778/0.745**	**0.426/0.392/0.434**	**0.776/0.820/0.801**

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TABLE IV
STATISTICAL RESULTS OF AVERAGE HV OBTAINED BY NSGA-II, MOEA/D, SPARSEEA, LSMOF, MOEA/PSL, CCMO, AND THE PROPOSED COE-DCDP ON THE ASSISTMENTS, JUNYI, AND SLP

Dataset	#Exercise (m)	NSGA-II	MOEA/D	SparseEA	LSMOF	CCMO	MOEA/PSL	CoE-DCDP
ASSISTments	50	0.4057	0.4591	0.5592	0.4475	0.4239	0.4817	0.5876
	100	0.5015	0.6042	0.6501	0.6197	0.5684	0.6258	0.8877
JunYi	30	0.5716	0.5988	0.7482	0.6717	0.6446	0.6519	0.7642
	50	0.7145	0.7242	0.8121	0.7746	0.7287	0.7894	0.8422
SLP	30	0.6962	0.6861	0.7831	0.7058	0.6823	0.6592	0.8004
	50	0.7905	0.8004	0.8455	0.8271	0.7793	0.7154	0.8819

Best result in each row is highlighted.

both three datasets, which further demonstrates the effectiveness of the proposed ENCD.

To sum up, the proposed ENCD is effective in not only diagnosing student's proficiency to assist our approach for the PEGA task but also cognitive diagnosis for answer prediction.

D. Effectiveness of the Proposed CoE-DCDP

As can be also found in Table II, despite the fact that same CD models (i.e., NeurealCD and ENCD) are used for MOEAs, the performance of CoE-DCDP and CoE-DCDP-NCD are significantly better than SparseEA-variants (SparseEA-NCD and SparseEA-ENCD), CCMO-variants (CCMO-NCD and CCMO-ENCD), and MOEA-Probs-variants (MOEA-Probs-NCD and MOEA-Probs-ENCD), which demonstrates that the proposed CoE-DCDP is a more effective evolutionary algorithm than SparseEA, CCMO, and MOEA-Probs to solve the CMOP.

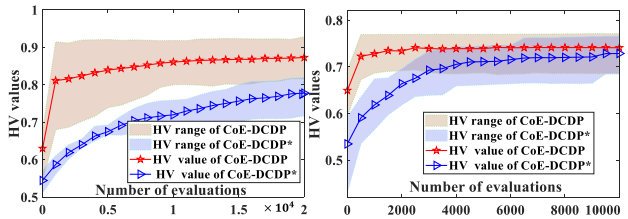
For a deeper insight into the effectiveness of the proposed CoE-DCDP, Table IV shows the comparison of the proposed CoE-DCDP and compared state-of-the-art MOEAs in terms of average hypervolume (HV) [53] values on ASSTISments, JunYi, and SLP, where the HV metric measures convergence and diversity of a population and a large HV value indicates a good convergence and diversity. Note that two settings about the number of recommended exercises m are used for fair comparison, where m is set to 50 and 100 for ASSISTments, and m is set to 30 and 50 for JunYi and SLP. As can be observed from Table IV, the proposed algorithm is superior to all comparison algorithms on three datasets under different settings for m. The effectiveness of the proposed CoE-DCDP is mainly attributed to the designed co-evolutionary framework that employs two populations equipped with two types of encoding, where the auxiliary population can accelerate the convergence of the main population to hold a better convergence and diversity than a single main population.

To demonstrate this, a CoE-DCDP variant (term CoE-DCDP*) is used for the subsequent comparison, where the CoE-DCDP* has only a main population without the help of the auxiliary population. For fast validation, the number of generations is set to 200 and 100 for ASSISTments, JunYi and SLP. Table V compares the HV values achieved by the CoE-DCDP and CoE-DCDP* on three datasets under two settings for m, it is obvious that HV values obtained by the proposed CoE-DCDP are better than that obtained by CoE-DCDP*, which indicates that better final convergence and diversity can be achieved with the help of the auxiliary population.

TABLE V
STATISTICAL RESULTS OF AVERAGE HV OBTAINED BY THE PROPOSED COE-DCDP AND ITS VARIANT ON THREE DATASETS, AVERAGED ON 100 STUDENTS

Dataset	# Exercise (m)	CoE-DCDP*	CoE-DCDP
ASSISTments	50	0.5103	0.5757
	100	0.7745	0.8732
JunYi	30	0.7367	0.7475
	50	0.7993	0.8246
SLP	30	0.7889	0.8004
	50	0.8616	0.8819

Best result in each row is highlighted.



(a) ASSISTments dataset (m=100) (b) JunYi dataset (m=30)

Fig. 9. Effectiveness verification of the auxiliary population in the proposed approach on two datasets. The red lines denote the results obtained by the proposed CoE-DCDP, while the blue lines are the results obtained by the CoE-DCDP without the auxiliary population.

To further investigate what specific effects the auxiliary population poses on the proposed CoE-DCDP, Fig. 9 depicts the convergence profiles of HV obtained by the two approaches on two datasets. Here only one setting for m is used to obtain the results of 100 students, where m is set to 100 and 30 for ASSISTments and JunYi. It is worth noting that the red and blue lines are HV convergence profiles averaged on 100 students (100 runs for each algorithm), and the light red and light blue areas denote the region of HV values. Two important facts can be observed: first, the proposed CoE-DCDP demonstrates high convergence speed at the early evolution stage, which proves that the auxiliary population indeed help the main population to accelerate its convergence. Second, the HV values of the CoE-DCDP are not only better but also more stable than that of the CoE-DCDP*, which indicates the devised co-evolutionary framework is more robust to different students with different abilities. Thus the auxiliary population can significantly accelerate and improve the convergence and diversity of the proposed CoE-DCDP.

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TABLE VI
STATISTICAL RESULTS OF AVERAGE HV OBTAINED BY THE PROPOSED COE-DCDP AND ITS VARIANTS ON THREE DATASETS

Method	ASSISTments	JunYi	SLP
CoE-DCDP[Initial](Random)	0.5432	0.5231	0.7217
CoE-DCDP[Initial]	0.6455	0.6154	0.8049
CoE-DCDP(R=0.1)	0.8375	0.7549	0.8716
CoE-DCDP(R=0.3)	0.8624	0.7544	0.8697
CoE-DCDP(R=0.5)	0.8602	0.7557	0.8719
CoE-DCDP(R=0.7)	0.8644	0.7555	0.8721
CoE-DCDP(R=0.9)	0.8559	0.7563	0.8722
CoE-DCDP(w/o local)	0.8086	0.7517	0.8594
CoE-DCDP	0.8877	0.7642	0.8819

Here, only the results under one setting is presented, where m is set to 100, 30, and 50 for assistments, JUNYI, and SLP.

To validate the effectiveness of other techniques, Table VI summarizes the comparison of the CoE-DCDP and its variants on three datasets under one setting, where CoE-DCDP[Initial] and CoE-DCDP[Initial](Random) refer to the initial results obtained by the proposed initialization method and the random initialization method, CoE-DCDP(R=0.1/0.3/0.5/0.7/0.9) denotes the result obtained by the CoE-DCDP with $Ratio$ fixed to 0.1/0.3/0.5/0.7/0.9, and CoE-DCDP(w/o local) indicates the result obtained by the CoE-DCDP without the local search strategy. As can be observed, the proposed initialization method, parameter adaptation strategy, and local search-based constraint repair strategy are effective in improving the proposed approach.

To summarize, not only the proposed CoE-DCDP is superior to state-of-the-art MOEAs, but also the employed auxiliary population are effective in improving the performance of CoE-DCDP.

VI. CONCLUSION AND FUTURE WORK

In this paper, we proposed the task of personalized exercise group assembly to flexibly recommend a suitable exercise group to a student with respect to her/his learning ability. To tackle the task, we designed three objectives and thus formulated the PEGA task as a CMOP. Then, an extended neural CD model was proposed to diagnose student's proficiency on all knowledge concept for computing the first objective, and a dual-encoding and dual-population based co-evolutionary algorithm was also proposed to solve the formulated constrained sparse large-scale MOP. Experimental results verified the effectiveness of exercises assembled by the proposed approach. In the future, we will be devoted to problem formulation and optimization approach design for various exercise group recommendation situations in actual learning process. Besides, we would like to extend our approach and further propose to solve other constrained sparse LSMOPs and expensive LSMOPs [54], [55].

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